Saturated Superfluid Vacuum Alpha

The Fine-Structure Constant as a Vortex Aspect Ratio

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Abstract

The fine-structure constant $\alpha \approx 1/137$ enters ordinary quantum electrodynamics as the dimensionless strength of electromagnetic coupling. In the Saturated Superfluid Vacuum (SSV) programme it appears more widely: as the strength of the chiral-shear sector, the ratio between the electron vortex-ring radius and the healing length, the particle-mass scale $\mu_0 = m_e/\alpha$, the small parameter controlling finite-ring corrections, and the geometric lever in weak-sector reconnection estimates. This paper isolates the common content of these appearances.

We do not claim here to derive the numerical value of α. Instead we show that, within SSV, the problem of deriving α is equivalent to a sharply defined defect-geometry problem: compute the equilibrium aspect ratio R^*/ξ of the stable toroidal electron vortex from the full LogSE plus chiral-shear functional, without inserting α as an empirical input, and show that

$$\frac{R^*}{\xi} = \alpha^{-1} \approx 137.036.$$

This reframes the fine-structure constant as a dimensionless eigenvalue of the vacuum defect spectrum rather than as a fundamental coupling constant postulated independently of the medium.

The paper also clarifies what this proposal is not. Following the propagation-speed audit in Paper II — since settled by the pre-registered dispersion computation of SSV issue 138, which shows the chiral-shear sector supports no propagating linear mode at all — α is not the ratio of the observed photon speed to the medium sound speed. The Goldstone/Bogoliubov channel at c is the remaining carrier candidate, but the photon mapping is not yet derived; α belongs to the static chiral-shear sector that fixes Coulomb strength and defect geometry. The central open calculation is therefore not a wave-speed calculation but a three-dimensional toroidal-vortex minimisation.

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1 Introduction

The fine-structure constant is one of the sharpest dimensionless numbers in physics:

$$\alpha = \frac{e^2}{4\pi\epsilon_0 \hbar c} \approx 7.29735256 \times 10^{-3}, \qquad \alpha^{-1} \approx 137.036. \tag{1}$$

In the Standard Model it is a measured low-energy coupling, whose running with scale is computed once the empirical value is fixed. Its numerical value is not derived from the structure of the theory.

The SSV programme changes the location of the question. In Paper I [1], particles were identified with topological defects of a saturated logarithmic superfluid: leptons as toroidal vortex rings and baryons as breather modes stabilised by knotted vortex skeletons. In Paper II [2], the force sector was developed and the earlier wave-speed interpretation of α was refined: the observed photon cannot be a chiral-shear mode propagating at $c_\perp$. A Goldstone/Bogoliubov excitation at c is the remaining carrier candidate, not yet a derived photon. The chiral-shear sector remains physically real, but its role is the static near-field Coulomb interaction and the geometry of charged defects.

This paper extracts the resulting implication:

> *In SSV, deriving the fine-structure constant is equivalent to deriving the aspect ratio of the stable toroidal electron vortex.*

The key target is

$$R^* = \frac{\xi}{\alpha}, \qquad \frac{R^*}{\xi} = \alpha^{-1}. \tag{2}$$

Here ξ is the electron healing length, the minor/core scale of the vortex ring, and R^* is the equilibrium major radius of the electron torus. Paper I used α as an empirical input to obtain $R^*/\xi \approx 137$. The present paper reverses the logic: the first principles calculation that remains to be done is the minimisation that outputs this number.

> **Claim of this paper.** The SSV framework does not yet derive α. It reduces the problem to a concrete eigenvalue calculation: solve the full 3D LogSE plus chiral-shear toroidal-vortex variational problem and compute the dimensionless equilibrium aspect ratio R^*/ξ. If the result is $137.036\ldots$, the fine-structure constant is geometrically derived as $\alpha = \xi/R^*$.

2 Inputs from the SSV Series

Only three pieces of the previous papers are needed here.

The saturated medium. The vacuum is described by a macroscopic complex order parameter $\Psi(\mathbf{x}, t)$. Its Madelung form gives a density and phase,

$$\Psi = \sqrt{\rho}\, e^{i\theta}, \qquad \mathbf{v} = \frac{\hbar}{m_0}\nabla\theta,$$

with quantised circulation

$$\oint \mathbf{v} \cdot d\mathbf{l} = n\kappa_0, \qquad \kappa_0 = \frac{h}{m_0}. \tag{3}$$

The logarithmic equation of state fixes the sound speed at saturation and the healing length

$$c_s(\rho_0) = c, \qquad \xi = \frac{\hbar}{\sqrt{2}\, m_0 c}. \tag{4}$$

The charged toroidal defect. The electron is identified with a stable toroidal vortex ring. Its minor/core radius is set by ξ; its major radius R is a variational degree of freedom. Charge is the chirality of the toroidal winding: the two signs of electric charge correspond to the two orientations of the vortex-ring phase winding.

The chiral-shear sector. Paper I introduced a chiral-shear contribution to the functional in order to make transverse, handed flow gradients energetic. Paper II clarified that the empirical α in this sector should be read as a dimensionless chiral-shear coupling, not as the speed of the observed photon. The Coulomb force is then recovered as a Bernoulli pressure of the chiral-shear near field,

$$F_C(r) = \frac{\alpha \hbar c}{r^2}, \tag{5}$$

after matching the elementary charge through $e^2/(4\pi\epsilon_0) = \alpha\hbar c$.

These inputs deliberately leave the numerical value of α open. That is the point of the present paper.

3 Where α Appears

The same dimensionless number appears in several places in the SSV series. The table below collects the appearances that matter for the derivation problem.

Sector	**Appearance of α**	**SSV reading**
Electromagnetic near field	$F_C = \alpha\hbar c/r^2$	chiral-shear coupling strength
Electron geometry	$R^*/\xi = \alpha^{-1}$	toroidal aspect ratio
Mass scale	$\mu_0 = m_e/\alpha$	flux-tube energy scale
Finite ring corrections	$(\xi/R^*)^2 = \alpha^2$	small expansion parameter
Weak reconnection estimates	$E_{\text{recon}} \sim m_e/\alpha^2$	ring-scale cavity energy
Gravity consistency	$G \propto \alpha^2/(N_p^2 m_e^2)$	mass-scale bookkeeping

This pattern is suggestive but not itself a derivation. The danger is to mistake recurrence for explanation. A number that appears everywhere in a framework may be fundamental, or it may be the shadow of one deeper geometric quantity. The SSV proposal is the latter: all appearances of α trace back to the relative scale of the electron torus and its core.

> **Reduction.** If the full toroidal-vortex calculation yields $R^*/\xi = 137.036\ldots$, then the Coulomb coupling, the mass scale μ_0, and the small expansion parameter all inherit the same number from one defect-geometry eigenvalue.

4 What α Is Not

The early SSV drafts treated α as a wave-speed ratio in the form $c_\perp/c$ or $c_\perp^2/c^2$, depending on convention. This was a useful scaffold because it placed α in the medium rather than in a separate gauge sector. But the propagation-speed audit in Paper II shows that this cannot be the final physical interpretation of the observed photon. Light in vacuum propagates at c, and so does gravitational radiation to the accuracy constrained by GW170817. The slower chiral-shear mode is therefore excluded as the carrier. The Goldstone/Bogoliubov channel at c is the remaining candidate, not a photon derivation by elimination.

This audit conclusion has since been replaced by a computation (SSV issue 138, CHIRAL-DISPERSION, pre-registered; see Paper II's resolved photon-speed section), and the computation is stronger than the audit: the chiral-shear sector supports *no propagating linear mode at all* around the uniform vacuum. The linear-order current perturbation is a pure gradient, so the $|\nabla \times \mathbf{j}|^2$ energy is quartic in the perturbation (measured exponent 4.000) and the λ-term is silent in the linear spectrum; the internal-direction (CP^1) channel disperses as $\omega = \hbar k^2/2m_0$ (measured power 2.000, coefficient the quantum-pressure scale), gapless, non-radiating, and decoupled from charge at linear order. There is no "slower second light" to reinterpret — the medium has exactly one signal speed, and α is the static stiffness of the defect sector. This also closes the would-be vacuum-Cherenkov objection: a propagating charge-coupled mode at αc would have made the vacuum dissipative for every charge above $\sim$14 eV-equivalent, excluded by synchrotrons and ultra-high-energy cosmic rays; no such mode exists in the linear spectrum.

This distinction is not a retreat from the SSV interpretation of α; it sharpens it. The role of α is not to slow photons. It is to set the strength and geometry of the chiral-shear sector:

$$\alpha \quad \text{belongs to coupling and defect geometry, not to photon speed.} \tag{6}$$

The time-dilation check in Paper II reinforces this split. In a density-depressed region near a mass, the longitudinal Bogoliubov speed scales as

$$\frac{c_s(\rho)}{c_s(\rho_0)} = \sqrt{\frac{\rho}{\rho_0}} \approx 1 + \frac{\Phi}{c^2}, \tag{7}$$

which has the correct weak-field sign for gravitational time dilation. A naive local-density reading of the chiral-shear speed has the opposite sign. The same audit therefore points to a clean separation:

$$c : \text{Goldstone/Bogoliubov propagation, light, gravity, clocks,} \tag{8}$$

$$\alpha : \text{chiral-shear coupling, Coulomb near field, defect aspect ratio.} \tag{9}$$

5 The Geometric Target

The electron ring is characterised by a major radius R and a minor/core radius a. The core minimisation is local and fixes

$$a^* = \xi. \tag{10}$$

The nontrivial calculation is the major-radius minimisation,

$$R^* = \arg\min_R E_{\text{torus}}[R, a^*;\, \mathcal{L}_{\text{LogSE}} + \mathcal{L}_\perp]. \tag{11}$$

The desired output is the dimensionless number

$$r^* \equiv \frac{R^*}{\xi}. \tag{12}$$

In Paper I, the empirical input α was used to write

$$r^* = \alpha^{-1}.$$

The alpha derivation problem is the inverse question:

$$r^*_{\text{computed}} \stackrel{?}{=} 137.036\ldots. \tag{13}$$

Open calculation. The alpha problem is not solved by restating $R^* = \xi/\alpha$. It is solved only if the functional minimisation computes R^*/ξ without inserting α anywhere as an empirical coupling or target radius.

6 Variational Form of the Problem

The most compact way to state the calculation is as a constrained minimisation over toroidal vortex configurations:

$$E[\Psi] = \int d^3x \left[\frac{\hbar^2}{2m_0} |\nabla\Psi|^2 + V_{\rm LogSE}(|\Psi|^2) + \mathcal{E}_\perp(\Psi, \nabla\Psi) \right], \tag{14}$$

subject to the topological constraints

$$\oint_{\rm poloidal} \nabla\theta \cdot d\mathbf{l} = 2\pi n, \qquad \oint_{\rm toroidal} \nabla\theta \cdot d\mathbf{l} = 2\pi m, \tag{15}$$

with the electron corresponding to the minimal charged toroidal sector. The chiral-shear term $\mathcal{E}_\perp$ must be written in a form that does not already contain the observed α as a numerical input. Its dimensionless coefficient should either be fixed by saturation/topology or absorbed into the same eigenvalue problem.

In the thin-ring language, the energy can be schematically decomposed as

$$E_{\rm torus}(R, a) = E_{\rm core}(a) + E_{\rm flow}(R, a) + E_{\rm shear}(R, a) + E_{\rm curvature}(R, a) + \cdots, \tag{16}$$

where

- $E_{\rm core}$ fixes the minor radius $a^* = \xi$;
- $E_{\rm flow}$ is the circulation energy of the vortex ring;
- $E_{\rm shear}$ is the chiral contribution that resists or favours expansion;
- $E_{\rm curvature}$ contains Kelvin, bending, and finite-core corrections.

The output of the calculation is not a mass in isolation, but the dimensionless stationary point

$$\frac{\partial E_{\rm torus}}{\partial R} = 0, \qquad \frac{\partial^2 E_{\rm torus}}{\partial R^2} > 0, \qquad r^* = \frac{R^*}{\xi}. \tag{17}$$

7 Avoiding the Circular Derivation

The most important discipline in an alpha paper is to avoid proving α by hiding it in the assumptions. The following routes are therefore disallowed as derivations:

(i) inserting $\lambda = \alpha^2\rho_0/m_0$ and then obtaining $R^*/\xi = \alpha^{-1}$;

(ii) imposing $e^2/(4\pi\epsilon_0) = \alpha\hbar c$ before solving the toroidal defect;

(iii) choosing a boundary condition equivalent to $R^*/\xi = 137$;

(iv) fitting a numerical coefficient in the shear term to recover the observed value.

A valid derivation must instead have the following form:

(D1) Write the LogSE plus chiral-shear functional with all dimensionless coefficients fixed by symmetry, saturation, or topology.

(D2) Solve the minimal charged toroidal defect in three dimensions.

(D3) Extract the equilibrium ratio R^*/ξ.

(D4) Only then identify $\alpha = \xi/R^*$ and compare with CODATA.

Success criterion. The alpha derivation succeeds if a calculation that contains no empirical α produces

$$R^*/\xi = 137.036\ldots$$

with a controlled error estimate. It fails if the result is not close, or if the value can be recovered only by inserting an equivalent dimensionless constant by hand.

8 Numerical Programme

The practical route is numerical before it is analytic. The cleanest calculation is a 3D constrained minimisation in the toroidal topological sector:

(N1) Choose a cylindrical or toroidal computational grid with adaptive refinement near the core.

(N2) Initialise Ψ with a minimal charged toroidal vortex of major radius R and core radius near ξ.

(N3) Evolve by imaginary time or gradient flow under the full LogSE plus chiral-shear functional.

(N4) Enforce the topological winding constraints so the defect cannot unwind.

(N5) Measure the relaxed R^*/ξ and its dependence on grid size, box size, and finite-core treatment.

(N6) Repeat across allowed chiral-shear functional variants to test whether 137 is robust or model-dependent.

The calculation has a clear falsification structure. A robust equilibrium near 137 would be strong evidence that α is indeed a vortex aspect ratio. A robust equilibrium at some unrelated number would falsify the specific SSV alpha proposal, even if other parts of the framework survived.

9 Consequences of a Successful Derivation

If $R^*/\xi = \alpha^{-1}$ is derived, several SSV claims become less independent than they currently appear.

Electromagnetic coupling. The Coulomb strength would be inherited from the geometry of the charged defect:

$$e^2/(4\pi\epsilon_0) = \hbar c\,\xi/R^*.$$

The electromagnetic coupling would no longer be a separately inserted number.

The mass scale. The scale $\mu_0 = m_e/\alpha$ would become

$$\mu_0 = m_e\,\frac{R^*}{\xi},$$

so the hadron mass scale would inherit the same factor from the electron-ring aspect ratio.

Weak-sector estimates. Quantities scaling as m_e/α^2 would become ring-geometry energies scaling as

$$m_e \left(\frac{R^*}{\xi}\right)^2.$$

The weak sector would still require the reconnection-cap calculation, but its small parameter would be geometrically anchored.

Perturbative corrections. Finite-ring corrections use

$$\varepsilon = (\xi/R^*)^2 = \alpha^2.$$

If α is geometric, the perturbation expansion is an expansion in the inverse aspect ratio of the electron torus.

10 Discussion

This paper is intentionally narrower than the force-sector unification of Paper II. Its purpose is to identify one load-bearing calculation. The fine-structure constant is not explained by showing that it appears in many places. It is explained only if all of those appearances trace back to a single dimensionless eigenvalue of the vacuum defect spectrum.

The connectedness of the SSV series is therefore not a weakness here. It is the reason the alpha problem is worth isolating. Coulomb strength, electron geometry, the mass scale μ_0, weak reconnection scales, and finite-ring corrections all point to the same ratio. If the ratio can be computed, the framework gains a central derivation. If it cannot, the failure is equally informative.

There is also an important conceptual payoff. The propagation-speed audit separates two roles that were previously entangled:

$$\text{propagation speed} \neq \text{coupling strength}.$$

The observed photon requires a propagation channel at c; the current SSV candidate is the Goldstone/Bogoliubov branch. The fine-structure constant belongs to the defect and near-field channel. This makes the alpha problem more geometric, not less.

11 Conclusion

The SSV framework does not yet derive the fine-structure constant. It does, however, sharpen the question. The fine-structure constant is not treated as an unexplained electromagnetic input, nor as a photon-speed ratio. It is the dimensionless chiral-shear/topological scale that fixes the electron torus relative to its healing length.

The derivation target is therefore:

$$\boxed{R^*/\xi = \alpha^{-1} \approx 137.036}$$

from the full 3D LogSE plus chiral-shear toroidal-vortex minimisation. That calculation is small enough to be stated precisely and large enough to matter. It is the point at which the SSV programme either turns the fine-structure constant into geometry, or learns exactly where the proposed geometry fails.

References

[1] S. Norland, *Saturated Superfluid Vacuum I: Topological Defects in a Saturated Superfluid Vacuum — The Geometric Origin of Matter*, 10.5281/zenodo.19651986, 2026.

[2] Norland, S., "Saturated Superfluid Vacuum II: Forces as Hydrodynamic Pressure Gradients in the Saturated Superfluid Plenum," SSV Series Paper II (2026).